Jiachen Yao Ph.D. Student

jiachen.yao@caltech.edu
https://www.jiachenyao.com/

Education

Ph.D. in Computing and Mathematical Sciences — Pasadena, California
California Institute of Technology — 2024 - 2029 *(expected)*

- Advisor: Professor Animashree Anandkumar
- Research area: Operator Learning and Generative Models

B.S. (Honours) in Information and Computer Science — Beijing, China
Zhili College, Tsinghua University — 2020 - 2024

- Advisor: Professor Jun Zhu
- Outstanding Graduates of Beijing

Publications

1. Xin Ju, **Jiachen Yao**, Anima Anandkumar, Sally M. Benson, and Gege Wen. Function-Space Decoupled Diffusion for Forward and Inverse Modeling in Carbon Capture and Storage. *ICLR Workshop on AI and PDE*, 2026. [link]
2. Thomas Lin*, **Jiachen Yao***, Lufang Chiang, Julius Berner, and Anima Anandkumar. Decoupled Diffusion Sampling for Inverse Problems on Function Spaces. *ICLR Workshop on AI and PDE*, 2026 (**Oral**). [link] [code]
3. Duy Nguyen*, **Jiachen Yao***, Jiayun Wang*, Julius Berner, and Anima Anandkumar. Flow-Guided Neural Operator for Self-Supervised Learning on Time Series Data. *Transactions on Machine Learning Research*, 2026. [link]
4. Bahareh Tolooshams*, Aditi Chandrashekar*, Rayhan Zirvi*, Abbas Mammadov, **Jiachen Yao**, Chuwei Wang, Anima Anandkumar. EquiReg: Equivariance Regularized Diffusion for Inverse Problems. *Transactions on Machine Learning Research*, 2026. [link]
5. **Jiachen Yao***, Abbas Mammadov*, Julius Berner, Gavin Kerrigan, Jong Chul Ye, Kamyar Azizzadenesheli, and Anima Anandkumar. Guided Diffusion Sampling on Function Spaces with Applications to PDEs. *NeurIPS*, 2025. [link] [poster] [code]
6. Zhongkai Hao*, **Jiachen Yao***, Chang Su*, Hang Su, Ziao Wang, Fanzhi Lu, ..., and Jun Zhu. PINNacle: A Comprehensive Benchmark of Physics-Informed Neural Networks for Solving PDEs. *NeurIPS*, 2024. [link] [poster] [code]
7. **Jiachen Yao***, Chang Su*, Zhongkai Hao, Songming Liu, Hang Su, and Jun Zhu. MultiAdam: Parameter-wise Scale-invariant Optimizer for Multiscale Training of Physics-informed Neural Networks. *ICML*, 2023. [link] [poster]
8. Yue Qin, Chun Yu, Wentao Yao, **Jiachen Yao**, Chen Liang, Yueting Weng, Yukang Yan, and Yuanchun Shi. Selecting Real-World Objects via User-Perspective Phone Occlusion. *CHI*, 2023. [link] [video]

$*$ denotes equal contributions.

Internships

Applied Science Intern | Amazon AWS — Jun 2026 - Sept 2026

- Time series foundation models with flow-based sampling.

Machine Learning Researcher | Subtle Computing — Jan 2024 - Aug 2024

- Developed an advanced voice recognition system for earbuds, enabling whisper transcription in noisy environments and challenging acoustic conditions.

Intern at Computer Science Department | Stanford University — June 2023 - Aug 2023

- Fully funded by Tsinghua University and Stanford University.
- Developed WhisperBuds, the first general-purpose, socially acceptable voice interface for AI agents on off-the-shelf earbuds. It achieved similar performance to traditional voice input while maintaining lower noise levels than most keyboards.

Awards and Honors

- Naren and Vinita Gupta Sensing to Intelligence Fellowship. 2026
- Professor Anatol Roshko Fellowship. 2025
- **Outstanding Graduates**, Beijing Municipal Commission of Education. 2024
- Scholarship for Innovation, Tsinghua University. 2023
- Selected to Spark Program at Tsinghua University, top 1%. 2023
- **National Scholarship**, Ministry of Education, PRC. 2022
- Scholarship for Comprehensive Excellence, Tsinghua University. 2021-2023
- **Silver Medal**, National Olympiad in Informatics. 2019

Skills

Hobbies: Badminton, Soccer; Blogging, Cooking, Movies.

Programming: Python, C/C++, JavaScript/TypeScript, MATLAB.

Tools: Git, CI/CD, Docker, Linux, Cloud & Local deployment.

Academic Services

Teaching:

Guest Lecturer, *Physics-Informed Machine Learning* (CEE 12-787), Carnegie Mellon University. 2026

Teaching Assistant, *Foundations of Machine Learning and Statistical Inference* (CS 165), Caltech; delivered one lecture. 2026

Reviewers for: *NeurIPS, ICML, ICLR, JMLR, TMLR, Nature Communications*.

Invited talks: Keynote for ICLR 2026 Workshop on AI & PDE, Brazil, 2026.
SIAM Conference on Uncertainty Quantification, Minneapolis, 2026.
AI & Scientific Discovery Workshop, UChicago, 2025.
Academia-Industry X Workshop, Caltech, 2024.